

Reinforcement Learning Laboratory

Lab Experiment 3

Title: Formulating Tic-Tac-Toe as an RL Problem

Duration: 3 Hours

CO 2: Decide if an application problem should be formulated as a RL problem; if yes be able to define it formally (in terms of the state space, action space, dynamics and reward model), state what algorithm (from class) is best suited for addressing it.

Students must formally define Tic-Tac-Toe as a reinforcement learning problem.

RL Components

Agent: The player (X or O).

Environment: Tic-Tac-Toe board and game rules.

State: Board configuration represented as a 3×3 grid.

Action: Placing a mark in an empty cell.

Reward: +1 for win, -1 for loss, 0 for draw or intermediate steps.

Episode Definition

An episode corresponds to one complete game of Tic-Tac-Toe. It starts from an empty board and ends when the agent wins, loses, or the game results in a draw. Episode length varies between 5 and 9 moves.

State Transition

At each time step, the agent observes the current state, selects an action, receives a reward, and transitions to the next state after the opponent's move. Transitions are deterministic and the environment is fully observable

Formulating Tic-Tac-Toe as a Reinforcement Learning Problem

RL Formulation

RL Element	Description
Agent	Player X (or O)
Environment	Tic-Tac-Toe board with game rules
State (S)	Current 3×3 board configuration
Action (A)	Place a mark in any empty cell
Reward (R)	+1 (Win), -1 (Loss), 0 (Draw/Intermediate)
Policy (π)	Strategy used to select actions
Value Function	Expected cumulative reward from a state

RL Element	Description
Episode	One complete game

State Representation

Represent the board using:

- **1** → Agent's move (X)
- **-1** → Opponent's move (O)
- **0** → Empty cell

Example

$$S = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

State Transition

At every time step:

1. Observe the board state.
2. Select an action.
3. Environment validates the move.
4. Opponent makes a move.
5. Receive reward.
6. Transition to the next state.

$$S_t \xrightarrow{A_t} S_{t+1}$$

Episode Definition

An episode begins with an empty board.

It terminates when:

- Agent wins
- Opponent wins
- Draw

Typical episode length:

$$5 \leq T \leq 9 \text{ moves}$$

Part C – Experimental Analysis

Observation Table

Experiment	Observation
Initial Behaviour	
Mid Training	
Final Behaviour	
Blocking Opponent	
Center Preference	
Corner Preference	
Draw Strategy	

Learning Curve

Students should plot

- Episodes vs Win Rate
- Episodes vs Draw Rate
- Episodes vs Average Reward

Expected Observations

Students should observe that:

- Initially, the agent plays randomly and frequently loses.
- After several hundred episodes, it begins recognizing immediate winning opportunities.
- The agent gradually learns to block opponent moves.
- It increasingly prefers the center and corner positions, which provide strategic advantages.
- As training continues, gameplay becomes more consistent and efficient.
- After sufficient training, the agent converges toward an optimal policy where losses become rare and most games between well-trained agents end in a draw.